

# ATISHAY JAIN

Software Engineering | Full-Stack Development | Applied AI

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## SUMMARY

Computer Engineering student with hands-on experience building production-oriented full-stack and AI systems using Python, FastAPI, React/Next.js, TypeScript, PostgreSQL, Docker, and modern LLM tooling. Built real-time web applications, secure backend workflows, offline AI pipelines, and CI/CD deployments, with a focus on reliability, performance, and iterative engineering.

## EXPERIENCE

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| <b>Def-Space Summer Intern - Bharat Space Education Research Centre (BSERC)</b>                                                                                                                                                                             | Jun 2026 - Aug 2026 |
| Def-Space Summer Internship Programme 2026   6+ weeks                                                                                                                                                                                                       |                     |
| <ul style="list-style-type: none"><li>Completed a structured Def-Space internship focused on defence-space innovation and responsible technology.</li></ul>                                                                                                 |                     |
| <b>Finance &amp; Documentation Manager - TEDxTIET</b>                                                                                                                                                                                                       | Aug 2025 - Feb 2026 |
| Patiala, Punjab, India                                                                                                                                                                                                                                      |                     |
| <ul style="list-style-type: none"><li>Managed a ₹3.5 lakh event budget with structured allocation, real-time expense tracking, variance analysis, and sponsor documentation, improving financial visibility and operational control across teams.</li></ul> |                     |

## PROJECTS

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| <b>ReceiptSplit - UPI-Native Bill Coordination Platform</b>                                                                                                                                                                                                                                                                                                                                      | 2026        |
| Next.js, TypeScript, FastAPI, PostgreSQL/Supabase, Docker                                                                                                                                                                                                                                                                                                                                        |             |
| <ul style="list-style-type: none"><li>Built bill coordination with receipt OCR, editable items, mixed splits, real-time state, and deterministic rounding.</li><li>Added auth, creator controls, settlement states, audit logging, UPI intents, and mobile-first failure-state UX.</li><li>Hardened permissions, status transitions, participant removal, and auditable dispute flows.</li></ul> |             |
| <b>TIET-LOOP - Campus Event Platform</b>                                                                                                                                                                                                                                                                                                                                                         | 2025 - 2026 |
| React, Vite, Tailwind CSS, FastAPI, WebSockets, Google Maps                                                                                                                                                                                                                                                                                                                                      |             |
| <ul style="list-style-type: none"><li>Built a campus PWA for events, RSVPs, chat, friendships, carpools, admin CRUD, and venue navigation.</li><li>Added REST/WebSocket flows, persistent storage, JWT auth, recommendations, and Google Maps.</li><li>Supported mobile-first access and event-management workflows for organizers.</li></ul>                                                    |             |
| <b>Hybrid GenAI Transaction Categorizer</b>                                                                                                                                                                                                                                                                                                                                                      | 2025 - 2026 |
| Python, FastAPI, React, ONNX Runtime, Qwen2.5-7B GGUF, SQLite                                                                                                                                                                                                                                                                                                                                    |             |
| <ul style="list-style-type: none"><li>Built an offline classifier combining merchant rules, ONNX DistilBERT, and local Qwen fallback.</li><li>Added feedback learning and React/FastAPI surfaces for human-readable predictions.</li><li>Tuned local CPU inference from roughly 60-80s to 5-13s through prompt/context tuning.</li></ul>                                                         |             |
| <b>Mahoraga - Adaptive RL Environment</b>                                                                                                                                                                                                                                                                                                                                                        | 2026        |
| Python, Gymnasium, FastAPI, Gradio, Qwen 2.5, LoRA                                                                                                                                                                                                                                                                                                                                               |             |
| <ul style="list-style-type: none"><li>Developed an adaptive RL environment with curriculum pressure and reward shaping.</li><li>Structured reward signals around survival, combat, adaptation, and anti-cowardice behavior.</li><li>Integrated Gymnasium, FastAPI/Gradio, and optional Qwen 2.5/LoRA paths for adaptive agent behavior.</li></ul>                                                |             |

## SKILLS

**Languages:** Python, C++, C, SQL, JavaScript/TypeScript, R  
**Full-Stack:** React, Next.js, Vite, Tailwind CSS, FastAPI, Flask, Django, REST APIs, WebSockets  
**Data & Infra:** PostgreSQL, Supabase, SQLite, MongoDB, Docker, Git, GitHub Actions, Linux, CI/CD  
**AI/ML:** Transformers, ONNX Runtime, local LLMs, RAG, embeddings, Pandas, NumPy

## EDUCATION & CERTIFICATIONS

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>B.Tech, Computer Engineering - Thapar Institute of Engineering and Technology</b>                                                                              | Aug 2023 - Jun 2027 |
| CGPA: 8   Patiala, Punjab, India                                                                                                                                  |                     |
| Class XII (Science) - Sacred Heart Senior Secondary School, Ludhiana   94.2%                                                                                      |                     |
| <b>Certifications:</b> NVIDIA Foundations of Deep Learning; Duke University - Introduction to RAG; CI/CD Pipeline with Docker; Software Development fundamentals. |                     |

## ACHIEVEMENT

Meta OpenEnv Hackathon 2026 Finalist - built Mahoraga, a self-improvement RL environment using Qwen2.5 and LoRA.